

一、判断题（每小题 2 分，共 20 分）

| 题号 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|----|---|---|---|---|---|---|---|---|---|----|
| 对错 | × | × | × | √ | × | √ | √ | × | × | √  |

二、填空题（每空 2 分，共 20 分）

1.  $\frac{1}{2}$       2.  $\underline{4}$       3.  $\underline{n-2}$       4.  $\underline{-\frac{1}{2}A}$      $\underline{-4}$
5.  $\underline{3}$       6.  $\underline{0}$      $\underline{(3, 4,}$       7.  $\underline{-1, 1, 5}$      $\underline{2}$

三、计算题（每小题 9 分，共 54 分）

$$1. \text{解: } D = \begin{vmatrix} 1 & 2 & 1 & 2 \\ -1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 0 \\ -1 & 2 & 5 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 1 & 2 \\ 0 & 2 & 2 & 4 \\ 0 & -3 & -1 & -4 \\ 0 & 4 & 6 & 6 \end{vmatrix} \dots\dots\dots 3'$$

$$= 1 \cdot \begin{vmatrix} 2 & 2 & 4 \\ -3 & -1 & -4 \\ 4 & 6 & 6 \end{vmatrix} = 2 \cdot \begin{vmatrix} 1 & 1 & 2 \\ -3 & -1 & -4 \\ 4 & 6 & 6 \end{vmatrix}$$

$$= 2 \cdot \begin{vmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ 0 & 2 & -2 \end{vmatrix} = -16. \blacksquare \dots\dots\dots 9'$$

$$2. \text{解: (1) 由于 } (A|E) = \left( \begin{array}{ccc|ccc} 2 & 3 & 2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & -1 & 0 & 0 & 1 \end{array} \right) \dots\dots\dots 3'$$

$$\rightarrow \left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -5 & -3 \\ 0 & 1 & 0 & -1 & 6 & 4 \\ 0 & 0 & 1 & 1 & -4 & -3 \end{array} \right)$$

$$\text{所以 } A^{-1} = \begin{pmatrix} 1 & -5 & -3 \\ -1 & 6 & 4 \\ 1 & -4 & -3 \end{pmatrix}. \blacksquare \dots\dots\dots 9'$$

$$3. \text{ 解: } \because (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{pmatrix} 4 & 0 & 4 & 1 \\ 1 & 2 & -9 & 0 \\ -1 & 3 & -16 & -1 \\ 2 & -4 & 22 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = (\beta_1, \beta_2, \beta_3, \beta_4)$$

$$\therefore r(\beta_1, \beta_2, \beta_3, \beta_4) \dots\dots\dots 5'$$

显然  $\beta_1, \beta_2, \beta_4$  是向量组  $\beta_1, \beta_2, \beta_3, \beta_4$  的一个极大线性无关组,

$$\text{且 } \beta_3 = \beta_1 - 5\beta_2 + 0\beta_4$$

故  $\alpha_1, \alpha_2, \alpha_4$  是  $\alpha_1, \alpha_2, \alpha_3, \alpha_4$  的一个极大线性无关组,

$$\text{且 } \alpha_3 = \alpha_1 - 5\alpha_2 + 0\alpha_4. \quad \blacksquare \quad \dots\dots\dots 9'$$

$$4. \text{ 解: } \text{由于 } (A|b) = \left( \begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & 1 & 0 \\ 0 & 1 & 2 & 2 & a \\ 5 & 4 & 3 & 3 & 2 \end{array} \right) \rightarrow \left( \begin{array}{cccc|c} 1 & 0 & -1 & -1 & -2 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & 0 & 0 & 0 & a-3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

所以当  $a=3$  时,  $r(A) = r(A|b) = 2 < 4$

从而原方程组有解, 且有无穷多解.  $\dots\dots\dots 5'$

$$\text{此时, 原方程组的同解方程组为: } \begin{cases} x_1 - x_3 - x_4 = -2 \\ x_2 + 2x_3 + 2x_4 = 3 \end{cases}$$

$$\text{令 } \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ 可得特解为: } \eta_0 = \begin{pmatrix} -2 \\ 3 \\ 0 \\ 0 \end{pmatrix}.$$

$$\text{导出方程组的同解方程组为: } \begin{cases} x_1 - x_3 - x_4 = 0 \\ x_2 + 2x_3 + 2x_4 = 0 \end{cases}$$

显然导出方程组有无穷多解, 自由未知量个数为 2,

取  $x_3, x_4$  为自由未知量, 令  $\begin{pmatrix} x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ , 可得导出组的基础解系为:

$$\xi_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \xi_2 = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix}.$$

故原方程组的通解为:

$$\eta = \eta_0 + k_1 \xi_1 + k_2 \xi_2$$

$$= \begin{pmatrix} -2 \\ 3 \\ 0 \\ 0 \end{pmatrix} + k_1 \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix}, \text{ 其中 } k_1, k_2 \text{ 任意. } \blacksquare \dots\dots\dots 9'$$

5. 解:

(1) 先正交化:

$$\text{令 } \beta_1 = \alpha_1 = (1, 1, 1, 1),$$

$$\beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1 = (1, -1, 0, 4) - \frac{4}{4}(1, 1, 1, 1) = (0, -2, -1, 3),$$

$$\begin{aligned} \beta_2 &= \alpha_3 - \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)} \beta_1 - \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)} \beta_2 \\ &= (3, 5, 1, -1) - \frac{8}{4}(1, 1, 1, 1) - \frac{-14}{14}(0, -2, -1, 3) = (1, 1, -2, 0). \quad \dots\dots\dots 6' \end{aligned}$$

(2) 再单位化:

$$\text{令 } \eta_1 = \frac{\beta_1}{|\beta_1|} = \frac{1}{2}(1, 1, 1, 1) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right),$$

$$\eta_2 = \frac{\beta_2}{|\beta_2|} = \frac{1}{\sqrt{14}}(0, -2, -1, 3) = \left(0, -\frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right),$$

$$\eta_3 = \frac{\beta_3}{|\beta_3|} = \frac{1}{\sqrt{6}}(1, 1, -2, 0) = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}, 0\right),$$

则  $\eta_1, \eta_2, \eta_3$  即为所求的标准正交向量组.  $\blacksquare \dots\dots\dots 9'$

6. 解: (1) 由于  $A$  相似于  $\Lambda$ , 所以  $tr(A) = tr(\Lambda)$ ,

$$\text{即 } 1 + 4 + a = 2 + 2 + 6,$$

$$\text{所以 } a = 5. \quad \dots\dots\dots 4'$$

(2) 显然  $\Lambda$  的特征值为: 2, 2, 6.

由于  $A$  相似于  $\Lambda$ , 所以  $A$  的特征值也为 2, 2, 6.

$$\text{当 } \lambda_1 = \lambda_2 = 2 \text{ 时, 解 } (2E - A)X = 0 \text{ 得基础解系为: } \xi_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \xi_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

当  $\lambda_3 = 6$  时, 解  $(6E - A)X = 0$  得基础解系为:  $\xi_3 = \begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ 1 \end{pmatrix}$ ;

故相似变换矩阵  $P = \begin{pmatrix} -1 & 1 & \frac{1}{3} \\ 1 & 0 & -\frac{2}{3} \\ 0 & 1 & 1 \end{pmatrix}$ . ■ .....9'

#### 四、证明题 (共 6 分)

证: 因为  $A^2 = A$ ,  $B^2 = B$ ,  $AB = BA$

所以  $(A + B - AB)^2$

$$= (A + B - AB) \cdot (A + B - AB) \quad \text{.....}2'$$

$$= A^2 + AB - A^2B + BA + B^2 - BAB - ABA - AB^2 + ABAB$$

$$= A + B - AB. \quad \blacksquare \quad \text{.....}6'$$